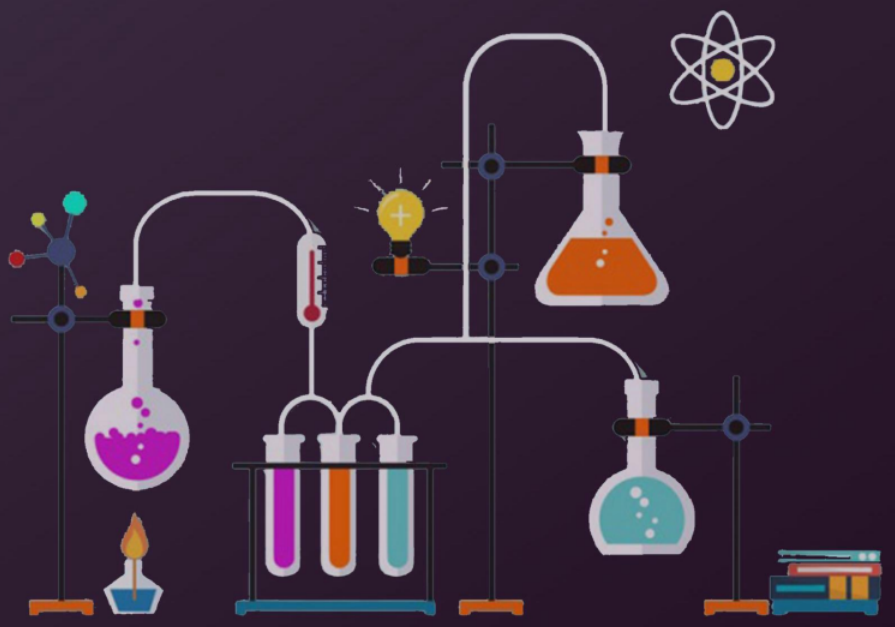
CHEMISTRY

Atomic Structure



SHEET

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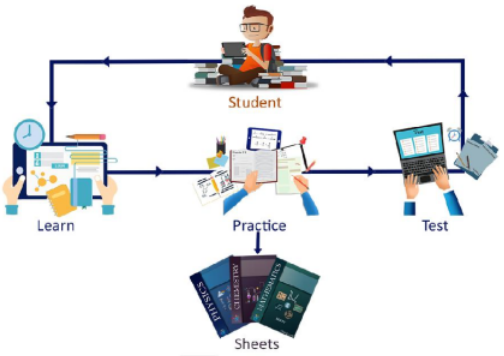
Contents

1.	Module Description	Page - ii
2.	Homework Index	Page - iii
3.	Exercise 1	Page - 1
4.	Exercise 1A	Page - 7
5.	Exercise 2	Page - 11
6.	Exercise 2A	Page - 14
7.	Answer Key	Page - 17

Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex 1	Ex 1A	Ex 2	Ex 2A
1	Discovery of electron and its charge	1-2	1		
2	Rutherford's model	3-7	2		
3	Size of Nucleus	8, 9	26-27, 32	1	
4	Wave theory	10-11, 13		2	
5	Planck's theory	12, 14-15	3, 28	3-4	3
6	Bohr's model	16-27	4, 5, 23	5-8, 10-11	1, 4-7, 17
7	Spectrum,Rydberg formula	28-34	6-7	12-14	8, 10, 15- 16
8	Photoelectric effect	35-39	9-10, 15-17		9, 12
9	Debroglie's hypothesis & Heisenberg's uncertainity principle	40-48	8, 11, 31	9, 15-16	11, 13
10	Quantum Numbers	49-54	12, 13, 18-22, 24, 29	17	14
11	Radial probability distribution function	55-58	25	18-22	2
12	Rules for filling of electron	59, 64		23-24	
13	Electronic configuration	60-63	30	25-28	
14	Higher element & Magnetic moment	65-66	14	29	
15	Basic concept of chemistry	67-69			

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Discovery of electron & it's charge

Q1. Which of the following properties of cathode rays is incorrect?

(A) Glow in the tube on application of current

(B) Defelection in Electric Field

(C) Flow from Anode to Cathode

(D) e/m ratio is independent of the cathode material

Q2. The minimum real charge on any Particle which can exist.

(A) 1.6×10^{-19}C (B) 1.6×10^{-10}C

(C) 4.8×10^{-10}C (D) Zero

Rutherford's model & size of nucleus

Q3. Which of the following statements are not true for anode ray ?

(A) They are deflected by electric and magnetic fields

(B) Their e/m ratio depends on the gas in discharge tube used to produce anode rays

(C) The e/m ratio of anode rays is constant.

(D) They are produced by ionization of the gas in discharge tube.

Q4. Rutherford's experiment, which established the nuclear model of atom, use a beam of ?

(A) β particles, which impinged on a metal foil and get absorbed

(B) γ-rays, which impinged on a metal foil and get absorbed.

(C) Helium atoms, which impinged on a metal foil and got scattered.

(D) Helium nuclei, which impinged on a metal foil and got scattered

Q5. In Rutherford's scattering experiment, which of the following does not happen?

(A) Most of the α-rays pass through without deflection.

(B) A few α-particles pass through the nucleus

(C) A few α-particles are deflected back.

(D) α-particles going near the nucleus are slightly deflected.

Q6. Rutherford α-particle scattering experiment led to the conclusion that ?

(A) mass and energy are related

(B) The mass and the positive charge of an atom are concentrated in nucleus.

(C) Neutrons are present in nucleus.

(D) atoms are electrically neutral

Q7. In Rutherford's scattering experiment an α-particle is projected with K.E. = 4 MeV, minimum distance upto which this α-particle can reach to the nucleus of Cu nucleus-

(atomic number of Cu = 29)

(A) 2.08×10^{-14} m

(B) 2.08×10^{-15} m

(C) 1.04×10^{-14} m

(D) 1.04×10^{-15} m

Q8. An atom having atomic mass number 13 has 7 neutrons. What is the atomic number of the atom?

(A) 6 (B) 7

(C) 8 (D) 9

Q9. An ion with mass number 56 contains 3 units of positive charge and 30.4% more neutrons than electrons. Assign the symbol to this ion.

(A) ${}^{56}_{18}Fe^{3+}$ (B) ${}^{56}_{22}Fe^{3+}$

(C) ${}^{56}_{26}Fe^{3+}$ (D) ${}^{56}_{28}Fe^{3+}$

Wave & Planck's Theory

Q10. Calculate the wave number of a radiation having wavelength 600 nm ?

(A) $1.67 \times 10^{8}\ m^{-1}$ (B) $1.67 \times 10^{10}\ m^{-1}$

(C) $1.67 \times 10^{12}\ m^{-1}$ (D) $1.67 \times 10^{6}\ m^{-1}$

Q11. The frequency of an electromagnetic radiation is 2×10^{6} Hz.

What is its wavelength in metres

(Velocity of light = $3 \times 10^{8}\ ms^{-1}$)

(A) 6.0×10^{14} (B) 1.5×10^{4}

(C) 1.5×10^{2} (D) 066×10^{-2}

Q12. Calculate the energy of a photon of sodium light of wavelength 2×10^{-16} m ?

(A) 9.9×10^{-10} J (B) 6.6×10^{-11} J

(C) 9.9×10^{-12} J (D) 4.8×10^{-11} J

Q13. The energies E_1 and E_2 of two radiations are 25 eV and 50eV respectively. The relation between their wavelengths i.e. λ_1 and λ_2 will be ?

(A) $\lambda_1 = \lambda_2$ (B) $\lambda_1 = 2\lambda_2$

(C) $\lambda_1 = 4\lambda_2$ (D) $\lambda_1 = \frac{1}{2}\lambda_2$

Q14. What is the number of photons of light with a wavelength of 4000 pm that provide 1 Joule of energy ?

(A) 5.4×10^{13} (B) 4.2×10^{14}

(C) 3.2×10^{8} (D) 2.0×10^{16}

Q15. The power output for a certain laser transition was found to be 2.79 J per sec per square meter. Given λ = 520 nm, calculate the number of quanta emitted per square meter per second.

(A) 3.10×10^{12} (B) 5.10×10^{12}

(C) 2.10×10^{18} (D) 7.30×10^{18}

Bohr Model

Q16. According to quantum restriction on angular momentum, imposed by Bohr :

(A) $mv = \frac{nh}{\pi}$

(B) $\lambda = \frac{h}{mv}$

(C) $mvr = \frac{nh}{2\pi}$

(D) $\bar{\upsilon} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

Q17. The distance between 2nd and 1st bohr orbit of hydrogen atom is ?

(A) 0.529×10^{-8} cm (B) 1.587×10^{-8} cm

(C) 2.116×10^{-8} cm (D) 1.058×10^{-8} cm

Q18. The ratio of the radii of the 5th orbits of He^+ & Li^+ will be?

(A) 2 : 3 (B) 3 : 2

(C) 4 : 1 (D) 5 : 3

Q19. The ratio of the radii of two Bohr orbit's of H-atoms is 4 : 9 what would be their nomenclature

(A) K and L (B) L and M

(C) N and L (D) 2 and 3 both

Q20. In the Bohr's orbit, what is the ratio of total potential energy and total energy of the electron?

(A) 1/2 (B) –2

(C) +1 (D) +2

Q21. Find out the K.E. of the first excited state of the hydrogen atom :-

(A) –13.6 ev (B) –3.4 ev

(C) + 3.4 ev (D) +13.6 ev

Q22. For which of the following species, Bohr's theory is not applicable ?

(A) Be^{+3} (B) Li^{+2}

(C) He^{+2} (D) H

Q23. Electron absorb photon when :-

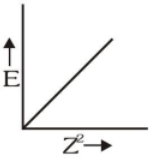
(A) it revolves in a stationary path

(B) it jumps from n = 2 to n = 1

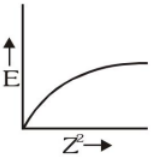
(C) it jumps from n = 1 to n = 2

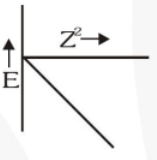
(D) all are correct

Q24. The graphical representation of energy of e^- and atomic number is ?

(A) 

(B) 

(C) 

(D) 

Q25. The ratio of $E_2–E_1$ to $E_4–E_3$ for the H-atom is approx. equal to ?

(A) 10 (B) 15

(C) 17 (D) 12

Q26. The ratio of the energy of electron in second excited state of He^+ ion to the energy of the electron in the first excited state of Be^{+3} is ?

(A) 1 : 3 (B) 9 : 16

(C) 1 : 9 (D) 16 : 9

Q27. If the speed of electron in the Bohr's first orbit of hydrogen atom is x, the speed of the electron in the Bohr's third orbit is ?

(A) x/9 (B) x/3

(C) 3x (D) 9x

Spectrum, Rydberg Formula

Q28. The line spectra of two elements are not identical because ?

(A) The elements do not have the same number of neutrons

(B) They have different mass number

(C) Their outermost electrons are at different energy levels

(D) They have different valencies

Q29. With a certain exciting radiation of a particular frequency, to which hydrogen atoms are exposed. the maximum number of spectral lines obtainable in the emission is 15. The uppermost energy level to which the electron is excited is n = ?

(A) 4 (B) 5

(C) 6 (D) 7

Q30. Calculate the wavelength (in Å) of radiation emitted, producing a line in Lyman series, when an electron falls from fourth stationary state to ground state in hydrogen atom.

(A) 970 Å (B) 900 Å

(C) 800 Å (D) 920 Å

Q31. If R is the Rydberg constant for hydrogen, then wave number of the first line in the Lyman series is ?

(A) 2R (B) $\frac{R}{2}$

(C) $\frac{3R}{4}$ (D) $\frac{R}{4}$

Q32. The wavelength of limiting line in balmer series of H-atom is ?

(A) 3.644 A° (B) 36.44 A°

(C) 364.4 A° (D) 3644 A°

Q33. Third line of Balmer series is produced by which transition in spectrum of H–atom ?

(A) 5 to 2

(B) 5 to 1

(C) 4 to 2

(D) 4 to 1

Q34. The longest λ for the Lyman series is ?

[Given R_H = 109678 cm^{-1}]

(A) 1215

(B) 1315

(C) 1415

(D) 1515

Photoelectric Effect

Q35. Which of the following is incorrect regarding photoelectric effect ?

(A) It helped in establishing the particle nature of light

(B) There is a minimum frequency below which no photoelectrons are ejected

(C) Photocurrent increases with increasing intensity

(D) Energy of all the photoelectrons ejected is the same

Q36. Which of the following explains almost instantaneous ejection of electrons from cathode?

(A) Energy of radiations being continuous

(B) Planck's idea of radiation being made of particles called quantum

(C) Low ionisation potential of cathode

(D) None of the above

Q37. In photoelectric effect work function of any metal is 2.5 eV. The most energetic emitted electron are stopped by potential of 1.5 Volt, then ?

(A) Energy of incident photons is 4eV

(B) Energy of incident photons is 1eV

(C) Wavelength of emitted electron is 1 Å

(D) Photocurrent increases with the application of photons of high frequency

Q38. The ejection of the photoelectron from the silver metal in the photoelectric effect experiment can be stopped by applying the voltage of 0.35 V when the radiation 256.7 nm is used. Calculate the work function for silver metal.

(A) 4.48 eV (B) 2.18 eV

(C) 3.18 eV (D) 0.21 eV

Q39. If the photon of the wavelength 150 pm strikes an atom and one of its inner bound electrons is ejected out with a velocity of 1.5×10^7 ms^{-1}, calculate the energy with which it is bound to the nucleus.

(A) 5.12×10^{-15} J (B) 1.22×10^{-15} J

(C) 4.34×10^{-14} J (D) 2.34×10^{-12} J

De Broglie's Hypothesis & Uncertainty Principle

Q40. Assuming that monsieur de-Broglie weighing 70 kg was jogging at 15 km/hr. Calculate the wavelength at which he was jogging.

(A) 4.166 m (B) 2.27×10^{-36} cm

(C) 2.27×10^{-36} m (D) None of these

Q41. Two particles of masses m and 2m have equal kinetic energies. Their de Broglie wavelengths are in the ratio of ?

(A) 1 : 1 (B) 1 : 2

(C) $1 : \sqrt{2}$ (D) $\sqrt{2} : 1$

Q42. The de Broglie wavelength associated with a ball of mass 1 kg having kinetic energy 0.5 J is ?

(A) 6.626×10^{-34} m

(B) 5.257×10^{-36} m

(C) 6.297×10^{-36} cm

(D) 7.287×10^{-36} cm

Q43. A beam of electron is accelerated by a potential difference of 100 volt. Find the wave length of wave associated with electron ?

(A) 1.23 Å (B) 12.3 Å

(C) 2.62 Å (D) 22.6 Å

Q44. If the de-Broglie wavelength of the fourth Bohr orbit of hydrogen atom is 4Å, the circumference of the orbit will be ?

(A) 4Å (B) 4 nm

(C) 16 Å (D) 16 nm

Q45. What is the minimum product of the uncertainty in position & the uncertainty in momentum of a moving electron ?

(A) $\frac{h}{4\pi}$ (B) $\frac{h}{2\pi}$

(C) $\frac{h}{3\pi}$ (D) $\frac{h}{\pi}$

Q46. If uncertainty in position of electron is zero the uncertainty in its momentum would be?

(A) Zero (B) $h/2\pi$

(C) $h/4\pi$ (D) Infinity

Q47. Uncertainty in position and momentum are equal. Uncertainty in velocity is ?

(A) $\sqrt{\frac{h}{\pi}}$ (B) $\frac{h}{2\pi}$

(C) $\frac{1}{2m}\sqrt{\frac{h}{\pi}}$ (D) None of these

Q48. In an atom, an electron is moving with a speed of 600 m/s with an accuracy of 0.005%. Certainty with which the position of the electron can be located is ?

($h = 6.6 \times 10^{-34}$ kg m^2s^{-1}, mass of electron, $e_m = 9.1 \times 10^{-31}$ kg)

(A) 1.52×10^{-4} m (B) 5.10×10^{-3} m

(C) 1.92×10^{-3} m (D) 3.84×10^{-3} m

Quantum Numbers

Q49. Which of the following is correct for 2p-orbitals ?

(A) $n = 1, \ell = 2$ (B) $n = 1, \ell = 0$

(C) $n = 2, \ell = 0$ (D) $n = 2, \ell = 1$

Q50. The correct set of quantum number of 3f orbital :

(A) 3, 3, 1, $\pm\frac{1}{2}$ (B) 3, 2, 2, $\pm\frac{1}{2}$

(C) 4, 3, 3, $\pm\frac{1}{2}$ (D) None of these

Q51. In the absence of external magnetic field f sub-shell has ?

(A) 5 degenerate orbitals

(B) 3 degenerate orbitals

(C) 7 degenerate orbitals

(D) Non-degenerate orbitals

Q52. Sub-shells present in N-shell will be ?

(A) 4s, 4p, 4d, 4f

(B) 3s, 3p, 3d

(C) 2s, 2p

(D) None

Q53. Which of the following is not permissible ?

(A) $n = 4, \ell = 3, m = 0$

(B) $n = 4, \ell = 2, m = 1$

(C) $n = 4, \ell = 4, m = 1$

(D) $n = 4, \ell = 0, m = 0$

Q54. An electron has spin quantum No. $+\frac{1}{2}$ and a magnetic quantum no. –1. It cannot be present in ?

(A) d-orbital (B) f-orbital

(C) s-orbital (D) p-orbital

Radial Probability Distribution Functions

Q55. The statements which is correct regarding Nodes ?

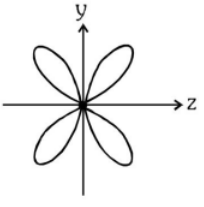
(A) Number of total node in an orbital = n

(B) Number of radial node in an orbital $= n - \ell - 1$

(C) Number of Angular node in an orbital = n-1

(D) All of the above

Q56. The nodal plane for the orbital given below is ?



(A) xy, yz (B) xz, yz

(C) xz, xy (D) Zero

Q57. The wave function of atomic orbital of H-like species is given as :

$$\psi_{2\sigma} = \frac{1}{4\sqrt{2\pi}} Z\frac{3}{2} (2-Zr)\, e^{-Zr/2}$$

The radius for nodal surface for He^+ ion in Å is ?

(A) 1.5 Å (B) 1 Å

(C) 2 Å (D) 4 Å

Q58. For an orbital in B^{+4} radial function is ?

$$R(r) = \frac{1}{9\sqrt{6}}\left(\frac{z}{a_0}\right)^{\frac{3}{4}} (4-\sigma)\sigma\, e^{-\sigma/2}$$

where $\sigma = \frac{Zr}{a_0}$ and $a_0 = 0.529$ Å ; Z = atomic number, r = radial distance from nucleus.

The radial node of orbital is at distance from nucleus.

(A) 0.529 Å (B) 2.12 Å

(C) 1.06 Å (D) 0.423 Å

Rules for Filling of Electrons & Electronic Configuration

Q59. The correct order of energy of subshells is ?

(A) 4f < 7s < 5d < 6p

(B) 4f < 5d < 6p < 7s

(C) 7s < 5d < 6p < 4f

(D) 5d < 6p < 4f < 7s

Q60. Arrange the following in increasing order of energy

(i) n = 4, ℓ = 2, m = –1, s = +½

(ii) n = 3, ℓ = 2, m = –1, s = –½

(iii) n = 4, ℓ = 0, m = 0, s = +½

(iv) n = 5, ℓ = 0, m = 0, s = –½

(A) (i) < (ii) < (iii) < (iv)

(B) (iii) < (ii) < (iv) < (i)

(C) (iii) < (iv) < (ii) < (i)

(D) (ii) < (iii) < (i) < (iv)

Q61. The four quantum numbers of the outermost orbital of K (atomic no. = 19) are ?

(A) n = 2, ℓ = 0, m = 0, s = $+\frac{1}{2}$

(B) n = 4, ℓ = 0, m = 0, s = $+\frac{1}{2}$

(C) n = 3, ℓ = 1, m = 1, s = $+\frac{1}{2}$

(D) n = 4, ℓ = 2, m = –1, s = $+\frac{1}{2}$

Q62. Correct set of four quantum numbers for valence electron of rubidium (Z = 37) is ?

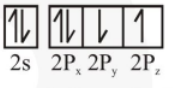
(A) 5, 0, 0, $+\frac{1}{2}$ (B) 5, 1, 0, $+\frac{1}{2}$

(C) 5, 1, 1, $+\frac{1}{2}$ (D) 6, 0, 0, $+\frac{1}{2}$

Q63. The electronic configuration of Fe^{+3} is ?

(A) $[Ar]3d^5\ 4s^1$ (B) $[Ar]3d^6\ 4s^0$

(C) $[Ar]3d^5\ 4s^0$ (D) $[Ar]3d^6\ 4s^2$

Q64.  this electronic arrangment can follow the hund's rule by _____ :-

(A) Changing the spin of electron present in the P_y orbital

(B) Changing the spin of electron present in the P_z-orbital

(C) Both (1) and (2) are correct

(D) Changing the spin of electron present in the P_x orbital

Q65. Maximum magnetic moment will be shown by ?

(A) $3d^5$ (B) $3d^2$

(C) $3d^9$ (D) $3d^7$

Q66. The spin only magnetic moment (in BM) of Cr^{+1} ion is ?

(A) $\sqrt{48}$ (B) $\sqrt{35}$

(C) $\sqrt{24}$ (D) $\sqrt{15}$

Basic Concepts of Chemistry

Q67. Select isoelectronic set :-

(a) K^+. S^{-2}, Sc^{+3}

(b) Na^+, H_3O^+, NH_4^+

(c) CO_2, H_3O^+, NH_4^+

(d) O_2^{+2}, N_2, C_2^{-2}

(A) a, c, d (B) b, c, d

(C) a, b, d (D) a, b, c

Q68. The charge of one mole of electrons is ?

(A) 3.64×10^2 C (B) 5.32×10^2 C

(C) 7.53×10^4 C (D) 9.63×10^4 C

Q69. Naturally occuring Neon has two isotopes whose atomic weight are 20 (I) and 22 (II). Atomic weight of natural neon is 20.2. The percentage of isotopes (I) and (II) respectively is ?

(A) 30 and 70 (B) 10 and 90

(C) 90 and 10 (D) 70 and 30

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following is not true for Thomson's model of the atom ?

(A) The radius of an electron can be calculated using this model

(B) In an undisturbed atom, the electrons will be embedded in positively charged spheres

(C) Positive charge is distributed evenly over the entire volume of the atom

(D) Electron revolve around the nucleus

Q2. Rutherford's experiment established that :

(A) Inside the atom there is a heavy positive centre

(B) Nucleus contains protons and neutrons

(C) Most of the space in the atoms is empty

(D) Size of the nucleus is very small

Q3. According to planck's quantum theory :

(A) Energy is absorbed or lost continuously in the form of quantum

(B) Energy is absorbed or lost dis-continuously in the form of quantum

(C) Electromagnetic radiations are made of small energy packets known as quantum

(D) Quantum, which has $\lambda = 3800\text{Å} - 7600\text{Å}$ is known as photon

Q4. Incorrect for Bohr's model :-

(A) applicable for any species

(B) $mvr = \frac{nh}{4\pi}$

(C) order of energies of orbitals $E_K < E_L < E_M$

(D) when electron jumps from one orbital to another energy is absorbed only

Q5. The ionization energy of hydrogen atom is 13.6 eV. The third ionization energy of lithium would be ?

(A) 16406 eV kJ/mol

(B) 11808 kJ/mol

(C) 40.8 eV

(D) 122.4 eV

Q6. In a He^+ sample electron make transition from 4th excited state to 2nd state then ?

(A) it can absorb or release energy

(B) 6 different spectral lines are observed

(C) number of lines belonging to the Balmer series is 3

(D) 10 different spectral lines are observed

Q7. Correct for EM waves :-

(A) have electrical field

(B) have magnetic field which is parallel to electrical field

(C) have mangetic field which is perpendicular to electrical field.

(D) velocity in vaccum is very less.

Q8. Which of the following relates to photons both as wave motion and as a stream of particles ?

(A) Interference (B) $E = mc^2$

(C) Diffraction (D) $E = h\upsilon$

Q9. Which statements is/are correct regarding the photoelectric effect -

(A) There is no electron ejection, regardless of intensity of the radiation unless the frequency, exceeds a threshold value which is the characteristic of metals.

(B) Maximum Kinetic energy of ejected electrons varies linearly with the frequency of incident radiation.

(C) Even at low intensity, electrons are ejected immediately if the frequency of incident light is more than the threshold frequency.

(D) An intense and weak beam of monochromatic radiation differs in having number photons and not in the energy of photons.

Q10. A monochromatic light of wavelength 155 nm is incident on a metal plate of work function 3.0 eV. Wavelength of emitted photo electrons may be ?

(A) $\sqrt{30}$ Å (B) $\sqrt{40}$ Å

(C) 6 Å (D) 5Å

Q11. Mathematical expression for uncertainity principle :-

(A) $\Delta x \cdot \Delta p = \frac{h}{4\pi}$ (B) $\Delta u \cdot \Delta x = \frac{h}{4\pi}$

(C) $\Delta x \cdot \Delta p = \frac{h}{4\pi m}$ (D) $\Delta v \cdot \Delta x = \frac{h}{4\pi m}$

Q12. True about quantum number :-

(A) ℓ gives idea of orientation

(B) n gives the size of an orbital

(C) s gives the direction of spin

(D) value of s can not be less than $-\frac{1}{2}$

Q13. Identify the correct statement(s) for principal quantum number (n = 4).

(A) maximum number of electron possible in this shell are 18

(B) Total number of subshell in this shell are 4

(C) Total number of electrons in the subshell possible with $n + \ell = 4$ can be 2.

(D) Total number of orbitals in this shell are 16

Q14. Identify the correct statement(s) –

(A) Angular momentum of electron in second (n = 2) Bohr's orbit will be 2.11×10^{-34} J-sec in any H-like atom.

(B) Spin only magnetic moment of any species is given by $\sqrt{n(n+2)}$ B.M. where n is the number of unpaired electrons in the species.

(C) Total spin of electron in a species can be calculated by the expression $\pm \frac{1}{2}n$ where n is the number of unpaired electrons in this species.

(D) A set of quantum number ($n = 3$, $\ell = 1$) is related with 3p orbital in any atom.

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for (Q. 15 to 17)

In the photoelectric effect the electrons are emitted instantaneously from a given metal plate, when it is irradiated with radiation of frequency equal to or greater than some minimum frequency, called the threshold frequency. According to planck's idea, light may be considered to be made up of discrete particles called photons. Each photon carries energy equal to hν. When this photon collides with the electron of the metal, the electron acquires energy equal to the photon.Thus the energy of the emitted electron is given by :

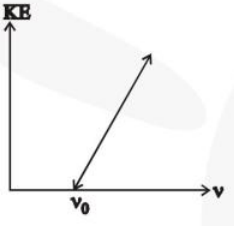
$$h\nu = KE_{maximum} + PE = \frac{1}{2}mv^2 + PE$$

If the incident radiation is of threshold frequency the electron will be emitted without any kinetic energy i.e.

$$\therefore \quad h\nu_0 = PE$$

$$\therefore \quad \frac{1}{2}mv^2 = h\nu - h\nu_0$$

A plot of kinetic energy of the emitted electron versus frequency of the incident radiation yields a straight line given as :-



Q15. A beam of white light is dispersed into its wavelength components by a Quartz prism and falls on a thin sheet of potassium metal. What is the correct decreasing order of kinetic energy of the electron emitted by the different light component –

(A) blue > green > orange > yellow

(B) orange > yellow > blue > violet

(C) yellow > green > blue > violet

(D) violet > blue > orange > red

Q16. A laser producing monochromatic lights of different wavelength is used to eject electron from the sheet of gold having threshold frequency $6.15 \times 10^{14}\ s^{-1}$ which of the following incident radiation will be suitable for the ejection of electrons ?

(A) One photon with frequency $5.16 \times 10^{15}\ s^{-1}$

(B) 1.5 moles of photons having frequency $3.05 \times 10^{14}\ s^{-1}$

(C) 0.5 moles of photon of frequency $12.3 \times 10^{12}\ s^{-1}$

(D) All of the above

Q17. The number of photo electrons emitted depends upon -

(A) The product of intensity and frequency of incident radiation

(B) The frequency of the incident radiation

(C) The intensity of the incident radiation

(D) None of these

Paragraph for (Q. 18 to 22)

Quantum numbers are required to specify the energy of electron, number of electron in any shell, to determine orbital angular momentum etc.Electrons are filled in the order of increasing energy starting from lowest energy. Any two electron in any atom can not have four quantum number same.

Q18. Maximum number of electron which may have $m = +1$ & $s = -\frac{1}{2}$ for Fe^{+2} :-

(A) 2 (B) 1 (C) 3 (D)4

Q19. Orbital angular momentum is non-zero for:-

(A) 1 s (B) 2 p (C) 3 s (D) 4 s

Q20. Which quantum number will affect if we remove one electron from Mg :-

(A) n (B) ℓ (C) m (D) s

Q21. Which quantum number represent the magnitude and direction of orbital angular momentum :-

(A) n, m (B) ℓ , m

(C) m, s (D) m, ℓ

Q22. In which of the following set/s of orbitals electron have equal orbital angular momentum :-

(A) 1 s, 2 p (B) 2 s, 2 p

(C) 2 p, 3 p (D) 3 p, 3 d

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question,choose the option corresponding to the correct matching.*

Q23. Match the column for the energy and radius of Hydrogen.

List I	**List II**
(A) Radius of 1st orbit	(P) – 13.6 eV/atom
(B) Radius of 2nd orbit	(Q) – 3.4 eV/atom
(C) Energy of 1st orbit	(R) 0.529 Å
(D) Energy of 2nd orbit	(S) 2.116 Å

	A	**B**	**C**	**D**
(A)	Q	R	P	S
(B)	S	R	Q	P
(C)	R	S	P	Q
(D)	S	R	P	Q

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q24. Match the column correctly –

Column I	**Column II**
(A) No. of orbitals in the nth shell	(P) $2(2\ell + 1)$
(B) Max. no. of electrons in a subshell	(Q) n
(C) No. of subshells in n^{th} shell	(R) $2\ell + 1$
(D) No. of orbitals in a subshell	(S) n^2

Q25. Match the column correctly –

Column I	**Column II**
(A) The radial node of 5s atomic orbital is	(P) 1
(B) The angular node of $3d_{yz}$ atomic orbital is	(Q) 4
(C) The sum of angular node and radial node of $4d_{xy}$ atomic orbital	(R) 2
(D) The angular node of 3p atomic orbital is	(S) 3

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q26. Atomic number of metal which used as foil in Rutherford experiment ?

Q27. α -particle has mass in amu ? [Write your Ans. nearest integer value]

Q28. Ratio of energies of a photon if its wave length is 6200 pm and 310Å is a/b. Find value of (a + b) ?

Q29. Maximum number of electrons possible in a sublevel is equal to a ℓ + b. Value of (a – b).

Q30. How many electron in $_{19}K$ have n = 3, ℓ = 0.

Q31. De-Broglie wavelength of a particle becomes $\frac{1}{\sqrt{2}}$ times when its K.E. increase x times. Value of x =?

Q32. If X^{+2} has 25 neutron, 28 electron, then calculate electron, proton & neutron in X^{-3}? [Write your answer as sum of electron + proton + nuetron]

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. What will be ratio of volume of nucleus with volume of atom ? Assume that nucleus and atom are spherical.

(A) 10^{-15} (B) 10^{15}
(C) 10^{5} (D) 10^{-5}

Q2. Which of the following statements is incorrect ?

(A) The frequency of radiation is inversely proportional to its wavelength
(B) Energy of radiation increases with increase in frequency
(C) Energy of radiation decreases with increases in wavelength
(D) The frequency of radiation is directly proportional to its wavelength

Q3. A bulb emits light of wavelength 4500Å. The bulb is rate as 150 watt and 8 percent of the energy is emitted as light. How many photons are emitted by the bulb per second?

(A) 15×10^{16} (B) 27.2×10^{18}
(C) 17.2×10^{16} (D) 25.2×10^{20}

Q4. The energy required to break one mole of Cl–Cl bonds in Cl_2 is 242 kJ mol^{-1}. The longest wavelength of light capable of breaking a single Cl–Cl bond is ?

($c = 3\times10^{8}\,ms^{-1}$ and $N_A = 6.02\times10^{23}\,mol^{-1}$)

(A) 494 nm (B) 594 nm
(C) 640 nm (D) 700 nm

Q5. The energy of an electron in excited hydrogen atom is –3.4 eV. Then, according to Bohr's theory the angular momentum of this electron in Jsec is ?

(A) 2.1×10^{-34} (B) 4.2×10^{-34}
(C) 6.6×10^{-34} (D) 3.2×10^{-33}

Q6. If radius of nth shell is 2.116 Å for H-atom calculate velocity of e^- in the same shell.

(A) 2.188×10^{6} m/s (B) 2.188×10^{12} m/s
(C) 1.094×10^{6} m/s (D) 1.094×10^{8} m/s

Q7. Which orbit of Be^{+3} has the same orbit radius as that of the ground state of hydrogen atom?

(A) 3 (B) 2
(C) 4 (D) 5

Q8. The energy required to dislodge electron from excited isolated H-atom, ($E_1 = -13.6$ eV) is?
(A) = 13.6eV
(B) >13.6eV
(C) <13.6 and >3.4eV
(D) $\leq$3.4 eV

Q9. The angular momentum of any electron can have the value(s) ?
(a) 3 (h/π) (b) 2.5 ($h/2\pi$)
(c) 0.5 (h/π) (d) (h/π)
(A) a, b, d (B) b, d
(C) a, b, c (D) a, c, d

Q10. The radii of any two of the first four Bohr's orbits of the hydrogen atom are in the ratio 1 : 4. The energy difference between them may be ?
(A) Either 12.09 eV or 3.4 eV
(B) Either 2.55 eV or 10.2 eV
(C) Either 13.6 eV or 3.4 eV
(D) Either 3.4 eV or 0.85 eV

Q11. The kinetic energy of an electron in second Bohr's orbit of a hydrogen atom is : (a_0 is Bohr's radius):-

(A) $\frac{h^2}{4\pi^2 m a_0^2}$ (B) $\frac{h^2}{16\pi^2 m a_0^2}$

(C) $\frac{h^2}{32\pi^2 m a_0^2}$ (D) $\frac{h^2}{64\pi^2 m a_0^2}$

Q12. If $\lambda = c_2\left[\frac{n^2}{n^2-2^2}\right]$ for balmer series, what is the value of c_2 ?

(A) $\frac{4}{R_H}$ (B) $\frac{2}{R_H}$

(C) $2R_H$ (D) $4R_H$

Q13. Which of the following statement(s) about spectral series is/are correct ?
(a) Paschen series appears in the infrared region.
(b) The lines in the Balmer series corresponds to electron transitions from higher energy levels to n = 2 energy level
(c) Transitions from higher energy levels to 5th energy level produce pfund series which falls in the infrared region
(d) The lines of Lyman series appear in the visible region
(A) a, b, d (B) b, d
(C) a, b, c (D) a, c, d

Q14. In hydrogen spectrum; If n = 4 then ratio of number of lines in emission spectrum, in the ultraviolet and visible regions is ?
(A) 2 : 3 (B) 3 : 1
(C) 3 : 2 (D) None of these

Q15. de Broglie equation has no significance for?
(A) Electron (B) Proton
(C) Neutron (D) Cricket ball

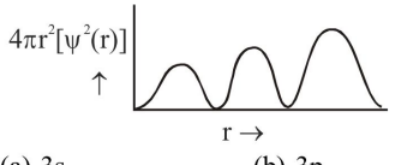
Q16. An α-particle is accelerated through a potential difference of V volts from rest. The de-Broglie's wavelength associated with it is ?

(A) $\sqrt{\frac{150}{V}}$ Å (B) $\frac{0.286}{\sqrt{V}}$ Å

(C) $\frac{0.101}{\sqrt{V}}$ Å (D) $\frac{0.983}{\sqrt{V}}$ Å

Q17. For an electron present in which of the following orbital for which $(n + l + m + s)$ value is maximum. Consider maximum possible value for 'm' and minimum possible value of 's' (where ever applicable).
(A) 3p (B) 5p
(C) 4d (D) 5s

Q18. This curve can show radial probability distribution of ?



(a) 3s (b) 3p
(c) 3d (d) 4p
(e) 5d
(A) a, b, c (B) b,d
(C) a,d,e (D) c,d,e

Q19. Which of the following statement is **FALSE?**

(A) Number of spherical node for 4d orbital is 3

(B) For p_y orbital xz plane is angular node

(C) For 's' orbital, ψ^2 is maximum at nucleus.

(D) Angular wave function of P_z orbital is proportional to cos q (Given : q is angle from z-axis)

Q20. The radial part of schrodinger wave equation for hydrogen atom is ?

$$\psi(r)=\frac{1}{16\sqrt{4}\,a_0^{3/2}}(\sigma-1)(\sigma^2-8\sigma+12)e^{-\sigma/2}$$

Where a_0 = constant & $\sigma = 2r/na_0$;

n = principle quantum number

Select the incorrect statement:

(A) Distance of nearest radial node from the nucleus is $2a_0$

(B) Distance of farthest radial node from the nucleus is $12a_0$

(C) $\psi(r)$ is for 4s orbital

(D) $\psi(r)$ is for 4p orbital

Q21. Find the group of orbitals in which atleast one common nodal plane is present :

(A) $4d_{x^2-y^2}, 5d_{z^2}$ (B) $2p_y, 3d_{xy}, 4d_{yz}$

(C) $2p_x, 2p_y, 4d_{yz}$ (D) $2p_z, 3d_{x^2-y^2}, 4d_{z^2}$

Q22. Which orbital notation does not have spherical node :-

(A) n = 2, $\ell = 0$ (B) n = 3, $\ell = 1$

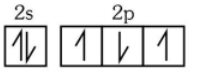
(C) n = 3, $\ell = 0$ (D) n = 1, $\ell = 0$

Q23. The orbital diagram in which aufbau principle is violated is ?

(A)  (B) 

(C) (D)

Q24. The orbital diagram in which Pauli exclusion principle, Hund's rule and Aufbau principle are violated is ?

(A) 

(B)

(C) 

(D)

Q25. Which of the following has the maximum number of unpaired electrons ?

(A) Mn (B) Ti

(C) V (D) Al

Q26. Predict the total spin in Ni^{+2} ion :-

(A) $\pm\frac{5}{2}$ (B) $\pm\frac{3}{2}$

(C) $\pm\frac{1}{2}$ (D) ± 1

Q27. Which of the following statement is true ?

(a) 6 electrons are present in Mg for which m = 0

(b) 6 electron are present in one p-orbital of Mg

(c) Maximum 18 electrons present in M-shell

(d) 3–electron present in phosphorous for which, $\ell = 0$, m = 0, s = $+\frac{1}{2}$

(A) a, b, c, d (B) a, b c

(C) a, c, d (D) c, d

Q28. Select the correct answer

(A) number of e^- in Cr for which $l = 0$ are 5

(B) two nodal plane present in $5p_x$ orbital

(C) $8e^-$ present in Mg for which $(n + l) = 3$

(D) All are correct

Q29. If spin only magnetic moment is 2.85 BM for x^{+2} then x is ?

(A) C (B) N

(C) O (D) F

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following statements (regarding an atom of H) is/are correct ?

(A) Kinetic energy of the electron is maximum in the first orbit

(B) Potential energy of the electron is maximum in the first orbit

(C) Radius of the second orbit is four times the radius of the first orbit

(D) Various energy levels are unequally spaced

Q2. For radial function & angular function correct option is/are?

(A) R(r) determine size of an orbital

(B) angular function determine energy

(C) R(r) contains n & m only

(D) q(q) contains ℓ & m.

Q3. Neon gas is generally used in the sign boards. If it emits strongly at 316 nm. Choose the correct options –

(A) The frequency of emission is 9.5×10^{14} Hz

(B) Distance travelled by this radiation in 30 s is 9×10^{6} km

(C) Energy of quantum is 6.29×10^{-19} J

(D) Number of quanta present if it produces 2J of energy is 3.18×10^{18}

Q4. The energy levels of hypothetical one electron atom are

– 0 eV	— $n = \infty$
– 0.50 eV	— $n = 5$
– 1.45 eV	— $n = 4$
– 3.08 eV	— $n = 3$
– 5.3 eV	— $n = 2$
– 15.6 eV	— $n = 1$

Choose the correct options

(A) The ionisation potential of atom is 15.6 eV

(B) The short wavelength limit of the series terminating at n = 2 is 233.9 nm

(C) The wave no. of photon emitted for the transition made by the electron from third orbit to its orbit is 1.009×10^{7} m^{-1}

(D) The ionisation potential of atom is 25.6 eV

Q5. Which is / are **correct** statement ?

(A) The difference in angular momentum associated with the electron present in consecutive orbits of H-atom is (n–1) $\frac{h}{2\pi}$.

(B) Energy difference between energy levels will be changed if, P.E. at infinity assigned value other than zero.

(C) Energy of the shells is in the order of **2 < 3 < 4**.

(D) On moving away from the nucleus, kinetic energy of electron decreases.

Q6. The I.P. of an atom is 16eV. What are wrong values of the value of first excitation potential?

(A) 10·2 eV (B) 12 eV

(C) 14 eV (D) 16 eV

Q7. 10.84×10^{23} hydrogen atoms are excited by radiations. The study of spectra indicates that 27% of the atoms are in 3rd energy level and 15% of atoms in 2nd energy level and the rest in ground state. IP of H is 13.6 eV. Choose the correct options –

(A) No. of atoms present in 1st energy level is 628.72×10^{21} atoms

(B) No. of atoms present in 2nd energy level 162.6×10^{21} atoms

(C) No. of atoms present in 3rd energy level 292.68×10^{21} atoms

(D) Total energy evolved when all the atoms return to ground state is 832.50 kJ

Q8. Which gives line spectrum ?
(A) Atomic Hydrogen
(B) Molecular Hydrogen
(C) Methane
(D) Atomic Helium

Q9. When photons of energy 4.25 eV strike the surface of a metal A, the ejected photoelectrons have maximum kinetic energy T_A (expressed in eV) and de Broglie wavelength λ_A. The maximum kinetic energy of photoelectrons liberated from another metal B by photons of energy 4.2 eV is $T_B = T_A - 1.50$ eV. If the de Broglie wavelength of these photoelectrons is $\lambda_B = 2\lambda_A$, then which is correct ?
(A) The work function of A is 2.25 eV
(B) The work function of B is 3.70 eV
(C) $T_A = 2.00$ eV
(D) $T_B = 2.75$ eV

Q10. A gas of identical H like atoms has some atoms in the lowest (ground) energy level A and some atoms in a particular upper (excited) energy level 'B' and there are no atoms in any other energy level. The atoms of the gas make transition to a higher energy level by ab sorbing monochromatic light of photon energy 2.7 eV. Subsequently, the atoms emit radiations of only six different photon energies. Some of the emitted photons have energy of 2.7 eV. Some have more and some less than 2.7 eV. Choose the correct options -
(A) The principal quantum number initially excited level B is 2
(B) The ionization energy for the gas atoms is 14.4 eV
(C) The maximum energy of the emitted photons is 13.5 eV
(D) The minimum energy of the emitted photons is 0.7 eV

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for (Q.11 to 13)

De Broglie proposed dual nature for electron by putting his famous equation $\lambda = \frac{h}{mv}$. Later on Heisenberg proposed uncertainty principle as $\Delta p.\Delta x \geq \frac{h}{4\pi}$. On the contrary particle nature of electron was established on the basis of photoelectric effect. When a photon strikes the metal surface, it gives up its energy to the electron. Part of this energy (say W) is used by the electrons to escape from the metal and the remaining energy imparts kinetic energy $\left(\frac{1}{2}mv^2\right)$ to the ejected photoelectron. The potential applied on the surface to reduce the velocity of photoelectron to zero is known as stopping potential.

Q11. Uncertainity in the position of an electron (mass 9.1×10^{-31} kg) moving with a velocity 300 m/s, accurate upto 0.001% will be ? $\left(\frac{h}{4\pi m_e} = 5.8\times10^{-5}\right)$
(A) 19.2×10^{-2} m (B) 5.76×10^{-2} m
(C) 3.84×10^{-2} m (D) 1.92×10^{-2} m

Q12. When a beam of photons of a particular energy was incident on a surface of a particular pure metal having work function = (40 eV), some emitted photoelectrons has stopping potential equal to 22V, some had 12V and rest has lower values. Calculate the wavelength of incident photons assuming that at least one photoelectron is ejected with maximum possible kinetic energy.
(A) 200 Å (B) 248Å
(C) 238Å (D) 310Å

Q13. The circumference of third orbit of H atom is 3nm. What may be the approximate wavelength of the photon required to just ionize electron from this orbit ?
(A) 91.1nm (B) 364.7nm
(C) 821nm (D) 205nm

TRIPLE MATCH COLUMN TYPE

*This section contains TWO tables (each having 3 column and 4 rows).Based on each table each question has FOUR options[A],[B],[C] AND [D].**ONLY ONE** of these four is correct.*

Column-I (Quantum No.)	**Column-II Quantises (e^-)**	**Column-III (Orbital)**
(I) n	(P) Direction of orbital angular momentum vector	(i) Size
(II) ℓ	(Q) Direction of spin	(ii) Orientation
(III) m	(R) Energy	(iii) None
(IV) s	(S) Magnitude of orbital angular momentum	(iv) Shape

Q14. Select the correct option :-

(A) II, P, iv (B) I, R, iv (C) III, P, ii (D) IV, Q, iii

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R), (S) and (T). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q15.

Column-I	**Column-II**
(A) $n = 6 \rightarrow n = 3$ (in H-atom)	(P) 10 lines in spectrum
(B) $n = 7 \rightarrow n = 3$ (in H-atom)	(Q) spectral lines in visible region
(C) $n = 5 \rightarrow n = 2$ (in H-atom)	(R) red line in spectrum
(D) $n = 6 \rightarrow n = 2$ (in H-atom)	(S) spectral lines in infra red region
	(T) 6 lines in spectrum

Q16. Match the column

Column I	**Column II**
(A) In a sample of H-atom for $5 \rightarrow 2$ transition	(P) Maximum 6 spectral lines will be observed from the sample
(B) In a sample of He^+ ion for $5 \rightarrow 1$ transition	(Q) A photon of wavelength $\lambda = \frac{36}{5R}$ can be emitted.
(C) In a sample of H-atom for $7 \rightarrow 3$ transition	(R) A photon of wave number $\bar{\nu} = \frac{7R}{144}$ can be mitted.
(D) In a sample of He^+ ion for $7 \rightarrow 4$ transition	(S) When single isolated atom/ion is considered, maximum 4 spectral lines are observed for given transition.

Q17. Match the column :

Column I	**Column II**
(A) Ratio of ionisation energy for He^+ ion and Be^{3+} ion, respectively	(P) 2 : 1
(B) Ratio of total energy of electron in 1st excited state of He^+ ion and potential energy of electron in 5th excited state of Li^{2+} ion, respectively	(Q) 4 : 1
(C) Ratio of time period of revolution of electron in 1st orbit of He^+ ion and 2nd orbit of Be^{3+} ion, respectively	(R) 1 : 4
(D) Ratio of radius of 3rd excited state of He^+ ion and first excited state of H-atom.	(S) 1 : 2

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
C	A	C	D	B	B	A	A	C	D
11	12	13	14	15	16	17	18	19	20
C	A	B	D	D	C	B	B	B	D
21	22	23	24	25	26	27	28	29	30
C	C	C	D	B	C	B	C	C	A
31	32	33	34	35	36	37	38	39	40
C	D	A	A	D	B	A	A	B	C
41	42	43	44	45	46	47	48	49	50
D	A	A	C	A	D	C	C	D	D
51	52	53	54	55	56	57	58	59	60
C	A	C	C	B	C	B	D	B	B
61	62	63	64	65	66	67	68	69	
B	A	C	C	A	B	C	D	C	

Ex-1A

1. A,D **2.** A,C,D **3.** B,C,D **4.** A,B,D **5.** D
6. B,C **7.** A,C **8.** B,D **9.** A,B,C,D **10.** A,B,C
11. A,D **12.** B,C **13.** B,C,D **14.** A,B,C,D **15.** D
16. A **17.** C **18.** C **19.** B **20.** D
21. B **22.** C **23.** (C)
24. A $\rightarrow$ S, B $\rightarrow$ P, C $\rightarrow$ Q, D $\rightarrow$ R **25.** A $\rightarrow$ Q , B $\rightarrow$ R , C $\rightarrow$ S, D $\rightarrow$ P
26. 79 **27.** 4 **28.** 6 **29.** 2 **30.** 2
31. 2 **32.** 88

Ex-2									
1	2	3	4	5	6	7	8	9	10
A	D	B	A	A	C	B	D	D	B
11	12	13	14	15	16	17	18	19	20
C	A	C	C	D	C	C	C	A	D
21	22	23	24	25	26	27	28	29	
B	D	B	C	A	D	D	C	C	

Ex-2A

1. A,C,D **2.** A, D **3.** A,B,C,D **4.** A,B,C **5.** C,D
6. A,C,D **7.** A,B,C,D **8.** A,D **9.** A,B,C **10.** A,B,C,D
11. D **12.** A **13.** C **14.** C
15. A $\rightarrow$ S,T; B $\rightarrow$ P,S; C $\rightarrow$ Q,T; D $\rightarrow$ P,Q,R
16. A $\rightarrow$ P,Q,R; B $\rightarrow$ S; C $\rightarrow$ R,S; D $\rightarrow$ P,Q
17. A $\rightarrow$ R; B $\rightarrow$ P; C $\rightarrow$ S; D $\rightarrow$ P

-: Important Notes & Formulas :-